Aleksas Mazeliauskas

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Professional experience

2022– Research group leader, **Institute for Theoretical Physics, Heidelberg University**

Six-year research project on the *Many-body QCD phenomena in high-energy proton and nuclear collisions* funded by DFG Emmy Noether Programme.

Principle investigator in the collaborative research center SFB 1225 *Isolated quantum systems and universality in extreme conditions*.

2019–2022 Senior research fellow, **Theoretical Physics Department, CERN**

Research on dense and hot QCD matter, Standard Model processes in nuclear environment, future ion experiments, Bayesian estimator of perturbative uncertainties `MiHO`.

2017–2019 Postdoctoral researcher, **Institute for Theoretical Physics, Heidelberg University**

Member of collaborative research center SFB 1225 *Isolated quantum systems and universality in extreme conditions*.

Work on non-equilibrium QCD physics using quark and gluon kinetic theory, pre-equilibrium kinetic propagator `KøMPøST`, resonance decay code `FastReso`.

2014–2017 Research assistant, **Nuclear Theory Group, Stony Brook University**

PhD dissertation work.

Education

2012–2017 **Stony Brook University**, United States, Doctor of Philosophy in Physics
Max Dresden Prize for outstanding theoretical thesis *Fluctuations in ultra−relativistic heavy ion collisions*, advisor Prof. Derek Teaney, degree awarded on 19/05/2017

2011–2012 **University of Cambridge**, United Kingdom, Master of Mathematics (Part III Maths)
graduated with distinction

2008–2011 **University of Cambridge**, United Kingdom, Bachelor of Arts (Mathematical Tripos)
1st class degree

Awards and distinctions

05/05/2022 6-year junior research group leader grant from the Emmy Noether Programme of German Research Foundation (DFG) (accepted)

22/02/2021 6-year junior researcher position from the Portuguese Foundation for Science and Technology (FCT) (declined)

09/11/2019 Nuclear Physics A Young Scientist Award for the best theory talk at Quark Matter 2019

02/05/2017 Max Dresden Prize for outstanding theoretical thesis

25/07/2016 APS FGSA Travel Award for Excellence in Graduate Research

23/05/2013 David Fox award for the outstanding Teaching Assistant

Professional services

Events organised

2022 Convener of the track *Event properties and hydro in small and large systems* at **Workshop on QCD Challenges from pp to AA**, 13-17 Feb 2023, Padova, Italy.

Co-organiser of the second edition of **The International Conference on Quantum Systems in Extreme Conditions (QSEC2022)**, 14-18 November 2022, Bingen, Germany.

Co-organiser of (virtual) **Second Lithuanian Particle Physics Meeting**, 11 April 2022.

2021 Co-organiser of (virtual) workshop on **Opportunities of OO and pO collisions at the LHC**, 4-10 Feb 2021 and the co-author of the summary report.

Co-organiser of (virtual) **First Lithuanian Particle Physics Meeting**, 9 March 2021.

2019 Convener of discussion group on *Connections between QGP-like observables in small systems* at **Workshop on QCD Challenges from pp to AA**, 19-23 Aug 2019, Lund, Sweden, and the editor of the summary report.

Convener of discussion group on *Implementations of critical dynamics* at **EMMI Rapid Reaction Task Force: Dynamics of critical fluctuations**, 8-12 April 2019, Darmstadt, Germany, and the editor of the summary report.

Refereeing

Physical Review Letters, Physical Review C and D, Physics Letters B, Nuclear Physics B, Journal of High Energy Physics

Teaching

Heidelberg University

2018-2019 **Co-Head tutor** for Quantum Field Theory and Advanced Quantum Field Theory courses. Responsible for preparing homework and exam problems, typesetting lecture notes and overseeing other tutors (winter and summer semesters).

2018 **Co-organiser** of a master level student lead seminar on QCD matter in heavy ion collisions (summer semester).

2017-2018 **Head tutor** for Quantum Field Theory course. Responsible for preparing homework, exam problems and overseeing nine other tutors (winter semester).

Stony Brook University

2014 **Recitation instructor** for calculus based undergraduate electromagnetism and mechanics courses (2 semesters). Selected student responses:

I think the instructor was more valuable than the course itself.

I would love to see him become a professor one day and attain his career goals.

He would do a fantastic job in teaching the entire course.

2012–2013 **Teaching assistant** for undergraduate mechanics laboratory (2 semesters), received David Fox award for outstanding Teaching Assistant.

National Student Academy (NMA)

2020– Head of Physics section at NMA – additional training school for gifted high-school children in Lithuania. Responsible for the distant learning program and in-person sessions.

2008–2011 Trained a number of young gifted Lithuanian students for physics competitions through distant learning programs and lecturing at summer schools.

Invited lectures

2021 two lectures at **CERN-Fermilab Hadron Collider Physics Summer School** (online) on *High-Density QCD with Proton & Ion Beams*, Geneva, Switzerland

Mentoring

2022-present PhD advisor for Fabian Zhou (Heidelberg University)

2020 Supervised a visiting PhD student Robin Törnkvist (Lund University)

2019 Informal mentoring of PhD students Oscar Garcia-Montero (Heidelberg University), Giuliano Giacalone (Université Paris Saclay) and Aleksandr Mikheev (Heidelberg University).

2018 Co-supervisor of a bachelor project by Paul Wiesemeyer (Heidelberg University) on radiative photon production in QGP.

Outreach

2022 Organised the display of *Forces of Nature* poster series celebrating women in Physics at CERN Theoretical Physics department.

2021 Conducted a virtual CMS tour and gave a talk about CERN during NMA winter session.

2020 Gave two talks to high-school children in Lithuania about science and work at CERN.

2019 Was interviewed about my work for French science magazine: J.-B. Veyrieras, *Découverte de la première flèche du temps*, Science & Vie, n°1228 (2020)

Prepared physics questions for an on-line science quiz competition *Išmanioji diena* held on 10/10/2019, Lithuania

2017 Prepared two introductory problem sets on elementary particles for high school students at Vilnius University Particle physics outreach group

2008-2013 One of founders and organisers of student-run physics team competition at Vilnius Lyceum, created a number of physics problems for the competition.

2007-2012 Active alumnus of NMA; spoke twice on Lithuanian national TV about education of talented children.

Languages

Lithuanian native, enthusiastic teacher

English fluent, higher education in English

German upper intermediate (B2)

French elementary (A2)

Russian basic conversational

Chinese beginner, passed HSK level II test

Programming

Programming Extensive numerically oriented coding in `C++` and `Python`, previous experience with `Fortran` and Matlab, good working knowledge of Mathematica

General IT Git (source code management), LaTeX (word processing), Inkscape (vector graphics), GNU/Linux (operating system), Gnuplot (graphing), HTML5 (web building)

Invited presentations

25/10/2022 talk at **Gluodynamics workshop** on *QCD thermalisation in kinetic theory*, Orsay, France

22/07/2022 talk at **Predictions for sPHENIX RBRC workshop (remote)** on *Jet and hadron nuclear modification factors*, BNL, United States

19/05/2022 talk at **X Large Hadron Collider Physics Conference (online)** on *Energy loss in small systems*, Taipei, Taiwan

13/05/2022 plenary talk at **Student Conference Mathematics and Natural Sciences: Theory and Application** on *Dalelių fizikos tyrimai CERN: naujų fundamentalių reiškinių ir dalelių paieškos*, Kaunas, Lithuania

25/11/2021 review talk at **III Italian Workshop about high energy physics with heavy ions** on *Theoretical overview on space-time evolution of heavy-ion collisions and QGP temperature*, Padova, Italy

31/08/2021 02/09/2021 two lectures at **CERN-Fermilab Hadron Collider Physics Summer School** (online) on *High-Density QCD with Proton & Ion Beams*, Geneva, Switzerland

30/07/2021 talk at **INT Program: Probing QCD at High Energy and Density with Jets** (online) on *Equilibration and collective effects in QCD kinetic theory*, Seattle, Washington, USA

27/07/2021 review talk at **European Physical Society Conference on High Energy Physics** (online) on *High-energy QCD Matter Theory*, Hamburg, Germany

28/06/2021 plenary talk at **Strong and Electro-Weak Matter Conference** (online) on *Matching glasma to hydro via kinetic theory*, Paris, France

03/03/2021 talk at **145th LHC Comittee Meeting** (online) on *Opportunities of OO and pO collisions at the LHC*, Geneva, Switzerland

15/01/2021 plenary talk at **VI Initial Stages Conference** (online) on *What do we learn from small systems about the physics of heavy ion collisions?*, Rehovot, Israel

24/09/2020 talk at **ALICE Physics Week (online)** on *Partonic rescattering in light nucleus collisions*, Bucharest, Hungary

28/05/2020 talk at **VIII Large Hadron Collider Physics Conference (online)** on *Equilibration of QGP in small systems*, Paris, France

06/12/2019 talk at **XIX Zimányi School Winter Workshop** on *Pre-equilibrium phenomena in Quark Gluon Plasma*, Budapest, Hungary

25/11/2019 talk at **Theoretical Foundations of Relativistic Hydrodynamics Workshop** on *Pre-scaling, hydrodynamic attractors and entropy production in heavy ion collisions*, Banff, Canada

01/11/2019 talk at **New Development of Hydrodynamics and its applications in Heavy-ion Collisions Workshop** on *Hydrodynamic attractors, initial state energy and particle production in relativistic nuclear collisions*, Shanghai, China

13/09/2019 talk at **XLIX International Symposium on Multiparticle Dynamics** on *Equilibration in Quark Gluon Plasma*, Santa Fe, New Mexico, USA

25/06/2019 plenary talk at **V Initial Stages Conference** on *Matching of initial conditions to hydro evolution*, New York, USA

06/04/2019 talk at **XIV Polish Workshop on Relativistic Heavy-Ion Collisions** on *Chemical equilibration in hadronic collisions*, Krakow, Poland

01/08/2018 parallel talk at **XIII Quark Confinement and the Hadron Spectrum Conference**, *2+1D simulations of pre-equilibrium stage with QCD kinetic theory*, Dublin, Ireland

18/05/2018 plenary talk at **XXVII Quark Matter Conference** on *Initial conditions for nuclear collisions: theory overview*, Venice, Italy

02/10/2017 talk at **XII High-pT Physics Workshop** on *Initial conditions for heavy ion collisions* , Bergen, Norway

15/06/2017 talk at **XII Workshop on Particle Correlations and Femtoscopy** on *Kinetic theory equilibration for realistic heavy ion initial conditions*, Amsterdam, The Netherlands

06/03/2015 talk at **Collectivity in Small Colliding Systems with High Multiplicity Workshop** on *Principal Component Analysis and Subleading Flow*, Brookhaven National Lab, New York, United States

Contributed presentations

02/12/2022 **QCD@LHC 2022 Conference**, Paris, France
06/04/2022 **XXIX Quark Matter Conference**, Krakow, Poland
07/10/2021 **XLIV National Lithuanian Physics Conference**, Vilnius, Lithuania
09/03/2021 **First Lithuanian Particle Physics Meeting** (online), Vilnius, Lithuania
03/06/2020 **X Hard Probes (online)**, Austin, USA
06/11/2019 **XXVIII Quark Matter Conference**, Wuhan, China, BEST THEORY TALK
22/09/2019 **Quantum Systems in Extreme Conditions**, Heidelberg, Germany
10/06/2019 **XVIII Strangness in Quark Matter**, Bari, Italy
14/01/2019 **XLVII Workshop on Gross Properties of Nuclei and Nuclear Excitations**, Hirschegg, Austria
23/05/2019 **Origins of Correlations in High Energy Collisions**, INT, Seattle, Washington, United States
09/04/2019 **EMMI RRTF: Dynamics of critical fluctuations**, Darmstadt, Germany
02/10/2018 **IX Hard Probes conference**, Aix-Les-Bains, France
27/08/2018 **Probing the Quark-Gluon Plasma with collective phenomena and heavy quarks**, MIAPP, Munich, Germany
19/09/2017 **IV Initial Stages**, Krakow, Poland
08/08/2017 **Critical Point and Onset of Deconfinement**, Stony Brook, New York, United States
07/02/2017 **XXVI Quark Matter**, Chicago, Illinois, United States
19/09/2016 **VII Hot Quarks**, South Padre Island, Texas, United States
25/05/2016 **III Initial Stages**, Lisbon, Portugal
22/07/2015 **Correlations and Fluctuations in p+A and A+A Collisions**, INT, Seattle, Washington, United States
27/03/2015 **Ohio-Region APS meeting**, Kent, Ohio, United States

List of publications

2022

1. M. Attems, J. Brewer, G. M. Innocenti, A. Mazeliauskas, S. Park, W. van der Schee and U. A. Wiedemann, *Medium-enhanced $c\bar{c}$ radiation*, arXiv:2209.13600
2. M. Attems, J. Brewer, G. M. Innocenti, A. Mazeliauskas, S. Park, W. van der Schee and U. A. Wiedemann, *The medium-modified $g \to c\bar{c}$ splitting function in the BDMPS-Z formalism*, arXiv:2203.11241
3. A. N. Mikheev, A. Mazeliauskas and J. Berges, *Stability analysis of non-thermal fixed points in longitudinally expanding kinetic theory*, **Phys. Rev. D** 105, 116025 (2022), arXiv:2203.02299

2021

4. W. Florkowski, A. Kumar, A. Mazeliauskas, R. Ryblewski, *Effect of thermal shear on longitudinal spin polarization in a thermal model*, **Phys. Rev. C** 105, 064901 (2022), arXiv:2112.02799
5. J. Brewer, A. Huss, A. Mazeliauskas, W. van der Schee, *Ratios of jet and hadron spectra at LHC energies: measuring high-p_T suppression without a pp reference*, **Phys.Rev.D** 105, 074040 (2022), arXiv:2108.13434

6. C. Duhr, A. Huss, A. Mazeliauskas, R. Szafron, *Analysis of Bayesian estimates for missing higher orders in perturbative calculations*, **JHEP** 09 (2021) 122, arXiv:2106.04585, https://doi.org/10.1007/JHEP09(2021)122
7. A. Andronic, P. Braun-Munzinger, M.K. Köhler, A. Mazeliauskas, K. Redlich, J. Stachel, V. Vislavicius, *The multiple-charm hierarchy in the statistical hadronization model*, **JHEP** 07 (2021) 035, arXiv:2104.12754, https://doi.org/10.1007/JHEP07(2021)035
8. A. Kurkela, A. Mazeliauskas, R. Törnkvist, *Collective flow in single-hit QCD kinetic theory*, **JHEP** 11 (2021) 216, arXiv:2104.08179, https://doi.org/10.1007/JHEP11(2021)216
9. J. Brewer, A. Mazeliauskas, W. van der Schee, *Opportunities of OO and pO collisions at the LHC*, arXiv:2103.01939

2020

10. A. Huss, A. Kurkela, A. Mazeliauskas, R. Paatelainen, W. van der Schee, U. Wiedemann, *Predicting parton energy loss in small collision systems*, **Phys.Rev.C** 103 054903 (2021) arXiv:2007.13758, https://doi.org/10.1103/PhysRevC.103.054903
11. A. Huss, A. Kurkela, A. Mazeliauskas, R. Paatelainen, W. van der Schee, U. Wiedemann, *Discovering partonic rescattering in light nucleus collisions*, **Phys.Rev.Lett.** 126 192301 (2021), arXiv:2007.13754, https://doi.org/10.1103/PhysRevLett.126.192301
12. J. Berges, M.P. Heller, A. Mazeliauskas, R. Venugopalan, *QCD thermalization: Ab initio approaches and interdisciplinary connections*, **Rev.Mod.Phys.** 93 (2021) 3 035003 arXiv:2005.12299, https://doi.org/10.1103/RevModPhys.93.035003
13. EDITOR FOR J. Adolfsson et al., *QCD Challenges from pp to A-A Collisions*, community report, **Eur.Phys.J.A 56, 11 (2020)**, arXiv:2003.10997
14. EDITOR FOR M. Bluhm et al., *Dynamics of critical fluctuations: Theory – phenomenology – heavy-ion collisions*, community report, **Nucl. Phys. A** 1003, 122016 (2002), arXiv:2001.08831

2019

15. O. Garcia-Montero, N. Löher, A. Mazeliauskas, J. Berges, and K. Reygers, *Probing the evolution of heavy-ion collisions using direct photon interferometry*, **Phys. Rev. C** 102, 024915 (2020), arXiv:1909:12246, https://doi.org/10.1103/PhysRevC.102.024915
16. D. Devetak, A. Dubla, S. Floerchinger, E. Grossi, A. Mazeliauskas, S. Masciocchi, and I. Selyuzhenkov, *Global fluid fits to identified particle transverse momentum spectra from heavy-ion collisions at the Large Hadron Collider*, **JHEP** 2006 (2020) 44 arXiv:1909:10485, https://doi.org/10.1007/JHEP06(2020)044
17. P. Hanus, A. Mazeliauskas, and K. Reygers, *Entropy production in pp and Pb-Pb collisions at energies available at the CERN Large Hadron Collider*, **Phys. Rev. C** 100, 064903 (2019), arXiv:1908.02792, https://doi.org/10.1103/PhysRevC.100.064903
18. G. Giacalone, A. Mazeliauskas and S. Schlichting, *Hydrodynamic Attractors, Initial State Energy, and Particle Production in Relativistic Nuclear Collisions*, **Phys. Rev. Lett.** 123, 262301 (2019), arXiv:1908.02866,https://doi.org/10.1103/PhysRevLett.123.262301
19. A. Mazeliauskas and V. Vislavicius, *Temperature and fluid velocity on the freeze-out surface from π, K, p spectra in pp, p–Pb and Pb–Pb collisions*, **Phys. Rev. C** 101, 014910 (2020), arXiv:1907.11059, https://doi.org/10.1103/PhysRevC.101.014910
20. W. Florkowski, A. Kumar, R. Ryblewski, and A. Mazeliauskas, *Longitudinal spin polarization in a thermal model*, **Phys. Rev. C 100**, 054907 (2019), arXiv:1904.00002, https://doi.org/10.1103/PhysRevC.100.054907

2018

21. A. Kurkela, A. Mazeliauskas, *Chemical equilibration in weakly coupled QCD*, **Phys. Rev. D** 99, 054018 (2019), EDITORS' SUGGESTION, arXiv:1811.03068, https://doi.org/10.1103/PhysRevD.99.054018
22. A. Kurkela, A. Mazeliauskas, *Chemical equilibration in hadronic collisions*, **Phys. Rev. Lett.** 122, 142301 (2019), arXiv:1811.03040, https://doi.org/10.1103/PhysRevLett.122.142301
23. A. Mazeliauskas, J. Berges, *Prescaling and far-from-equilibrium hydrodynamics in the quark-gluon plasma*, **Phys. Rev. Lett.** 122, 122301 (2019), arXiv:1810.10554, https://doi.org/10.1103/PhysRevLett.122.122301
24. A. Mazeliauskas, S. Floerchinger, E. Grossi, D. Teaney, *Fast resonance decays in nuclear collisions*, **Eur. Phys. J. C** (2019), arXiv:1809.11049, https://doi.org/10.1140/epjc/s10052-019-6791-7
25. A. Kurkela, A. Mazeliauskas, J.-F. Paquet, S. Schlichting and D. Teaney, *Matching the non-equilibrium initial stage of heavy ion collisions to hydrodynamics with QCD kinetic theory*, **Phys. Rev. Lett.** 122, 122302 (2019), EDITORS' SUGGESTION, arXiv:1805.01604, https://doi.org/10.1103/PhysRevLett.122.122302
26. A. Kurkela, A. Mazeliauskas, J.-F. Paquet, S. Schlichting and D. Teaney, *Effective kinetic description of event-by-event pre-equilibrium dynamics in high-energy heavy-ion collisions*, **Phys. Rev. C,99**, 034910 (2019), arXiv:1805.00961, https://doi.org/10.1103/PhysRevC.99.034910

pre-2018

27. Y. Akamatsu, A. Mazeliauskas and D. Teaney, *Bulk viscosity from hydrodynamic fluctuations with relativistic hydro-kinetic theory*, **Phys. Rev. C** 97, 024902 (2018), arXiv:1708.05657, https://doi.org/10.1103/PhysRevC.97.024902,
28. Y. Akamatsu, A. Mazeliauskas and D. Teaney, *A kinetic regime of hydrodynamic fluctuations and long time tails for a Bjorken expansion*, **Phys. Rev. C**,95, 014909 (2017), arXiv:1606.07742 https://doi.org/10.1103/PhysRevC.95.014909
29. L. Keegan, A. Kurkela, A. Mazeliauskas and D. Teaney, *Initial condition for hydrodynamics from weakly coupled pre-equilibrium evolution*, **JHEP** 1608 (2016) 171, arXiv:1605.04287, https://doi.org/10.1007/JHEP08(2016)171
30. A. Mazeliauskas and D. Teaney, *Fluctuations of harmonic and radial flow in heavy ion collisions with principal components*, **Phys. Rev. C** 93, 024913 (2016), arXiv:1509.07492, https://doi.org/10.1103/PhysRevC.93.024913
31. A. Mazeliauskas and D. Teaney, *Subleading harmonic flows in hydrodynamic simulations of heavy ion collisions*, **Phys. Rev. C** 91, 044902 (2015), arXiv:1501.03138, https://doi.org/10.1103/PhysRevC.91.044902

Selected Conference proceedings

1. A. Mazeliauskas, **Quark Matter 2018** *Initial conditions for nuclear collisions: theory overview*, Nucl. Phys. A 982 (2019), 134-141, arXiv:1807.05586